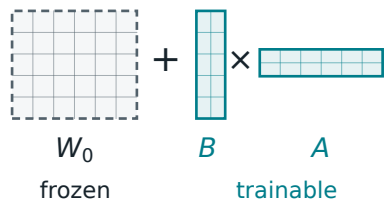


LoRA: adapt a frozen model with low-rank updates

a Task and idea

Adapt a language model to a downstream task.

Learn a low-rank weight update



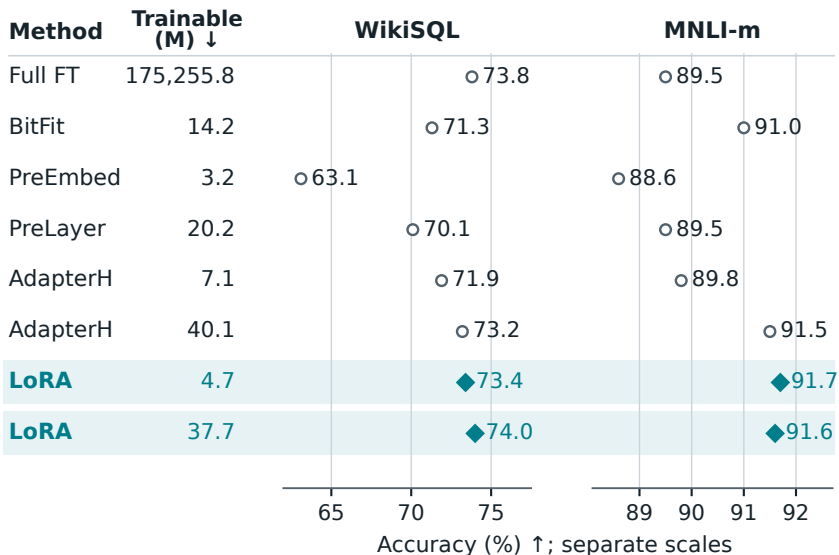
$$h = W_0 x + (\alpha/r) B(Ax)$$

NL-to-SQL example (schematic)

How many teams?

→ `SELECT COUNT(*) FROM teams`

b GPT-3 175B: reported validation results



4.7M LoRA vs full FT: -0.4 pp WikiSQL; +2.2 pp MNLI-m

Table 4, Hu et al. (2021), v2. Typical seed SD: WikiSQL $\approx$ 0.5 pp; MNLI-m $\approx$ 0.1 pp. No per-entry intervals.